



MATERI HANYA DIGUNAKAN TERBATAS UNTUK PERKULIAHAN
TI5003 PEMODELAN DAN BERPIKIR SISTEM

TI5003 PEMODELAN DAN BERPIKIR SISTEM

3. PROBLEM FORMULATION & SSM

LUCIA DIAWATI

EG INDUSTRIAL SYSTEM AND TECHNO-ECONOMY

FACULTY OF INDUSTRIAL TECHNOLOGY

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LEARNING OUTCOMES

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Mahasiswa mampu:

1. menjelaskan situasi masalah
2. memformulasikan masalah serta mengidentifikasi elemen dan *stakeholders* masalah
3. Memahami dan menggunakan *soft system methodology* untuk merepresentasikan dan memformulasikan masalah.



1. PROBLEM – SYSTEM – METHODOLOGY

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Continuum of Managerial Problematic Situations:

Well Structured



Unstructured

System Continuum:

“Hard Systems”
Thinking
“**OPTIMIZING**”



“Soft Systems”
Thinking
“**LEARNING**”

Methodology Continuum:

**Hard Systems
Methodology**

- System Analysis
- System Engineering
- Operation Research



System Dynamics

**Soft Systems
Methodology**

- Soft System
Methodology
- Interactive Planning



2. HARD AND SOFT SYSTEM THINKING

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1. “Hard System” Thinking

- Assumes:
 - **Objective reality of systems in the world**
 - **Well-defined problem** to be solved
 - **Technical factors** foremost
 - **Scientific approach to problem-solving**
 - **An ideal solution**
- HARD systems - rigid techniques and procedures to provide unambiguous solutions to **well-defined data** and **processing problems problems**, focused on computer implementations:



2. HARD AND SOFT SYSTEM THINKING

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2. “Soft System” Thinking

- Assumes:
 - **organisational problems are ‘messy’ or poorly defined**
 - **stakeholders interpret problems differently** (no objective reality)
 - **human factors important**
 - **creative, intuitive approach to problem-solving**
 - outcomes are **learning, better understanding**, rather than a ‘solution’
- Engineering approach can be inappropriate for ‘soft problems’ (with fuzzy requirements).
- SOFT systems - a loose framework of tools to be used at the discretion of the analyst, focused on improvements to organisational problems.



2. HARD AND SOFT SYSTEM THINKING

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1. **Hard System Methodology / Hard OR → operational decision making**

- Operation Research
- System Engineering
- System Analysis
- System Dynamics

2. **Soft System Methodology / Soft OR → Problem Structuring Methods (PSM) → strategic decision making**

- Checkland's **Soft Systems Methodology**
- Eden's **Strategic Option Development And Analysis (SODA)**
- Friend and Hickling's Strategic Choice Approach.

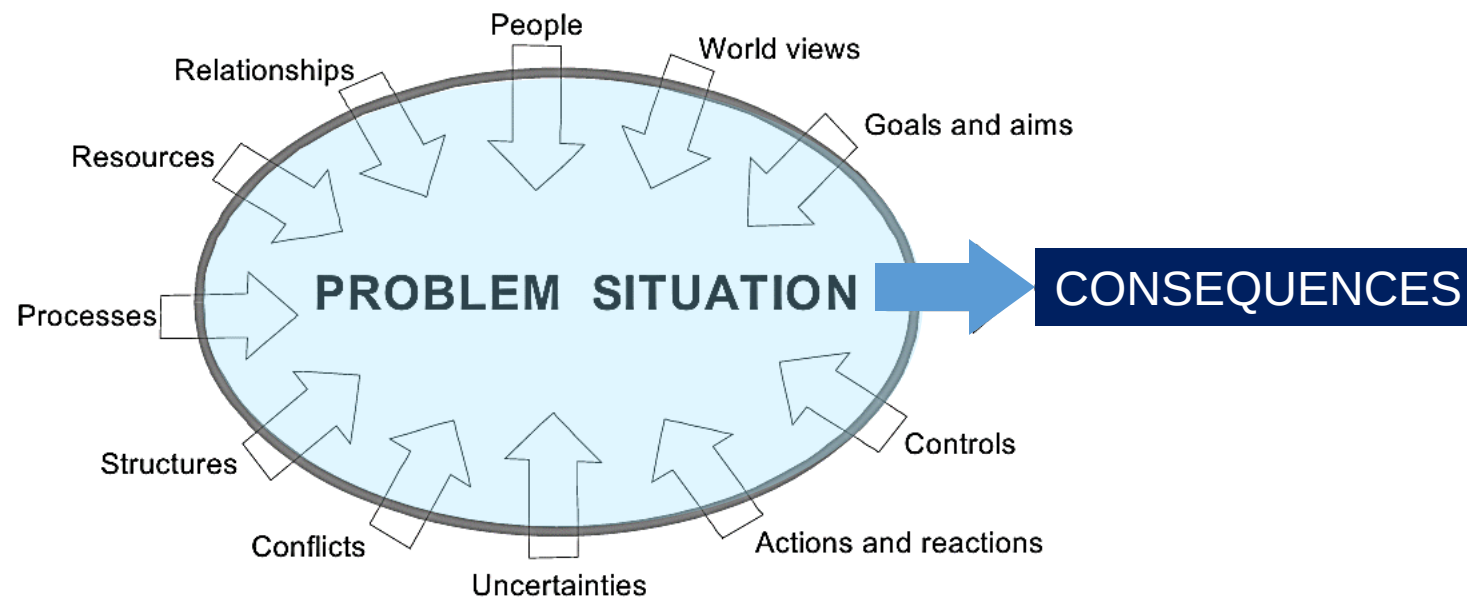


3. PROBLEM SITUATION, PROBLEM & ROOT CAUSE

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1. Problem Situation

- The context within which the problem occurs.
- The sum or aggregate of all aspects that can or may affect or shape the problem or issue of concern





3. PROBLEM SITUATION, PROBLEM & ROOT CAUSE

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Identifikasi Masalah (Problem):

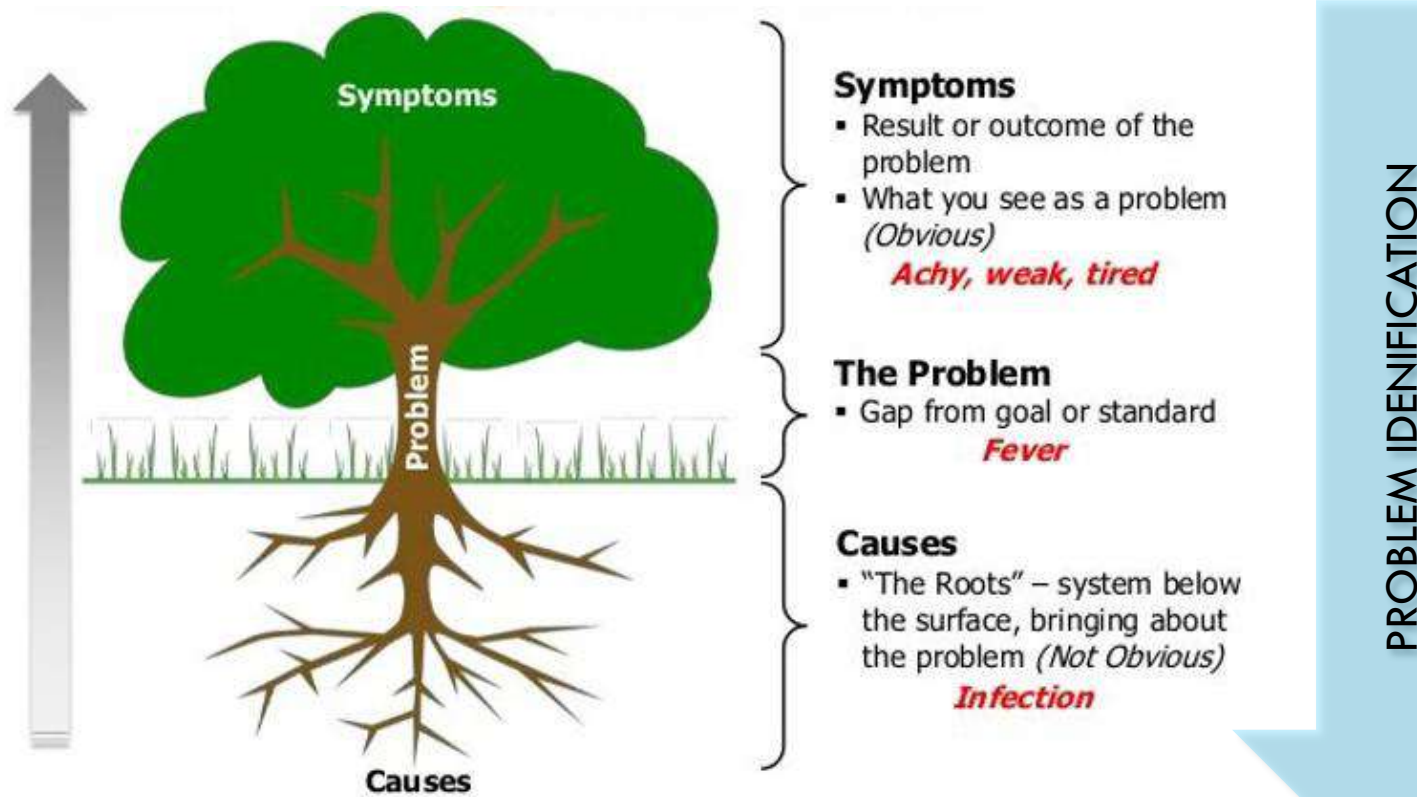
- Merupakan dasar pemodelan
- Akrabkan diri dengan situasi masalah:
 - Situasi
 - Proses
 - Struktur
 - Orang yang terlibat
 - Tujuan dan keinginan orang-orang
 - Hubungan antar orang-orang yang terlibat
 - Hirarki/struktur kekuasaan
 - Sumber daya yang tersedia
 - Sumber data & informasi



3. PROBLEM SITUATION, SYMPTOM AND PROBLEM

2. Symptom, Problem and Root Causes

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- Symptom: “apa yang tampak”
- Problem adalah “apa yang terjadi”
- Causes adalah “mengapa hal itu terjadi.”

UNDERSTANDING ROOT CAUSES

Modified from
<https://leantech.no/en/leansixsigma-blog/61-root-cause-analysis>



3. PROBLEM SITUATION, SYMPTOM AND PROBLEM

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Problem

The existence of a problem refers to **problem owner** (or a group of individuals) — usually its decision maker — who:

- is **dissatisfied** with the current state of affairs within a real-life context, or has some unsatisfied present or future needs, i.e. has some goals or objectives to be achieved or targets to be met;
- is **capable of judging** when these goals, objectives, or targets have been met to a satisfactory degree; and
- **has control over some aspects** of the problem situation that affect the extent to which goals, objectives, or targets can be achieved.



3. PROBLEM SITUATION, SYMPTOM AND PROBLEM

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Six Elements of Problem

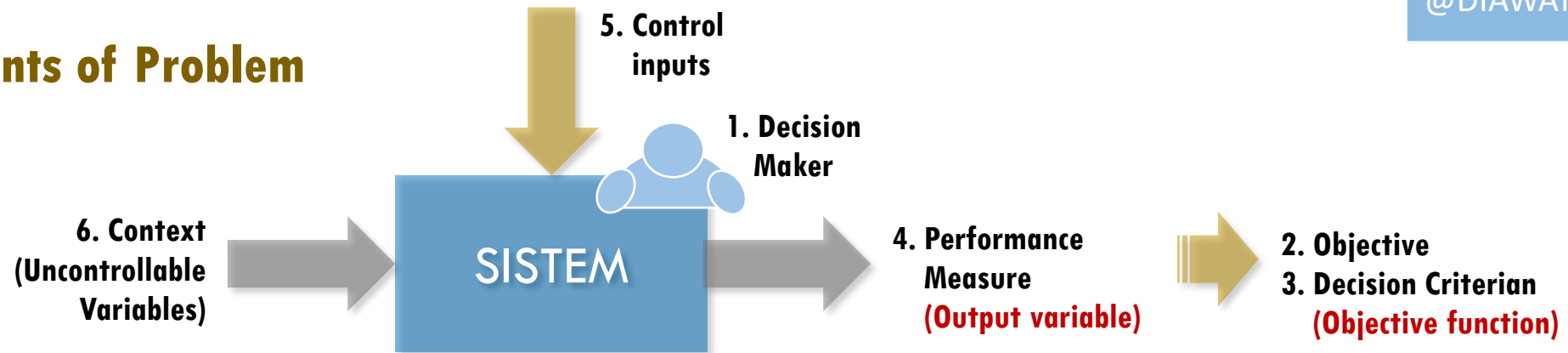
1. **Decision maker:** problem's owner
2. **Decision maker's objectives:** goals or objectives to be achieved or targets to be met
3. **Decision criterion:** the value process or standard used for judging or evaluating which decision or solution should be selected.
4. **Performance measure:** System output used for **measuring the system performance** as a function of the control inputs (**output variable**).
5. **Control inputs or alternative courses of action:** an input into a system over which the **decision maker has control** in the form of: **a discrete action** or **setting the values of decision variables** (**controllable input variable**).
6. **Context in which the problem occurs:** all aspects that directly or indirectly **affect the measure of performance** and over which the **decision maker has no immediate control** or which are taken as 'givens' (**uncontrollable input variable**).



3. PROBLEM SITUATION, SYMPTOM AND PROBLEM

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Six Elements of Problem



1. **Decision maker:** LOD Manager
2. **Objective:** achieving low operating cost for the LOD's operation
3. **Decision criterion:** minimizing total costs → **Objective function**
4. **Performance measure:** total operating costs of the LOD system → **Output variable**
5. **Control inputs:**
 - 1) Size of stock replenishment batches and
 - 2) Cutoff point for classifying customer orders as big or small
6. **Context or wider system of interest** (uncontrollable inputs): the LOD operation and the refinery



3. PROBLEM SITUATION, SYMPTOM AND PROBLEM

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Contoh: 6 Elemen Masalah dalam Linear Programming (LP)

1. **Decision maker** (Pengambil Keputusan):

- Pihak yang bertanggung jawab atas keputusan produksi.
- Contoh: Manajer produksi atau kepala pabrik.

2. **Decision maker's objectives** (Tujuan pengambil keputusan):

- **Target yang ingin dicapai.**
- Dalam kasus ini: **Meminimumkan total biaya produksi** sambil tetap memenuhi permintaan pasar.

3. **Decision criterion** (Kriteria Keputusan):

- **Standar untuk memilih** alternatif terbaik.
- Contoh: **Total biaya produksi** dibandingkan dengan kapasitas dan permintaan.
- Alternatif yang dipilih adalah yang menghasilkan biaya paling rendah namun *feasible*.



3. PROBLEM SITUATION, SYMPTOM AND PROBLEM

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4. **Performance measure** (Ukuran Kinerja):

- **Indikator kuantitatif** yang menunjukkan hasil keputusan.
- Contoh: 1) Total biaya produksi (Rp); 2) Jumlah unit yang diproduksi sesuai permintaan; 3) Efisiensi penggunaan bahan baku dan jam kerja mesin.

5. **Control inputs or alternative courses of action** (Variabel keputusan / alternatif tindakan):

- Faktor yang bisa dikendalikan oleh pengambil keputusan.
- Contoh: Jumlah unit produk A, B, C yang diproduksi.

6. **Context** (Konteks Masalah):

- Faktor eksternal yang tidak bisa langsung dikendalikan, tetapi memengaruhi hasil.
- Contoh: 1) Permintaan pasar (harus dipenuhi); 2) Kapasitas mesin dan tenaga kerja; 3) Harga bahan baku dari supplier; 4) Regulasi industri (misalnya standar kualitas atau batas emisi).



3. PROBLEM SITUATION, SYMPTOM AND PROBLEM

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Ringkasan:

1. **Decision Maker** → Manajer Produksi
2. **Objectives** → meminimumkan biaya produksi.
3. **Criterion** → biaya total sebagai dasar memilih solusi.
4. **Performance measure** → angka biaya (jumlah unit, efisiensi).
5. **Control inputs** → jumlah produk yang diproduksi (variabel keputusan LP).
6. **Context** → permintaan, kapasitas, harga bahan baku (kendala LP).



4. STAKEHOLDERS OR ROLES OF PEOPLE IN SYSTEMS

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1. **Problem owners:**

- the persons who have control over certain aspects of the problem situation, in particular over the choice of action to be taken.
- Most often, they are also the decision makers.

2. **Problem users (solution users)**

- The persons who use the solution and/or execute the decisions approved by the problem owners or decision makers.
- They have no authority to change the decision or initiate new action.

3. **Problem customers:** the beneficiaries or victims of the consequences of using the solution.

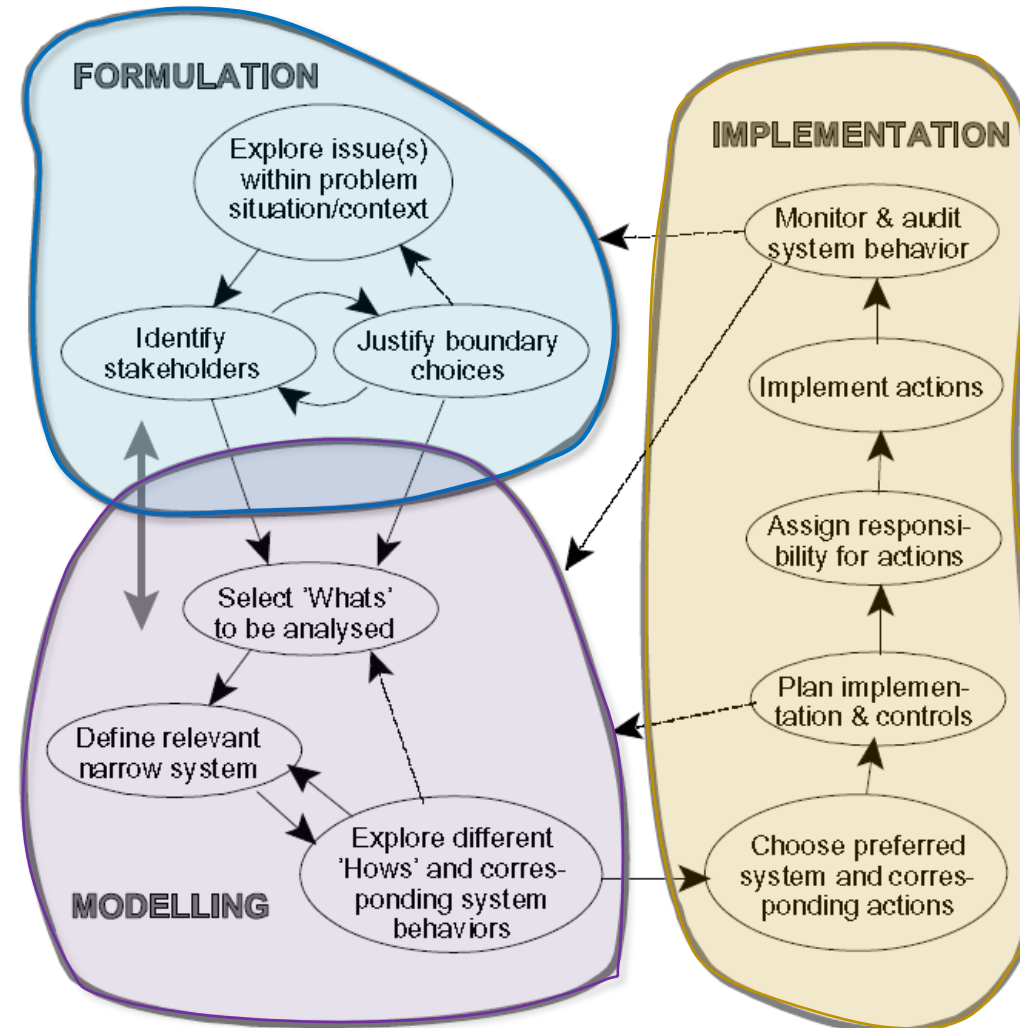
4. **Problem analysts or solvers:** the persons who analyse the problem and develop a solution for approval by the problem owners.



5. PROBLEM STRUCTURING METHOD

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General Working Mode of PSMs



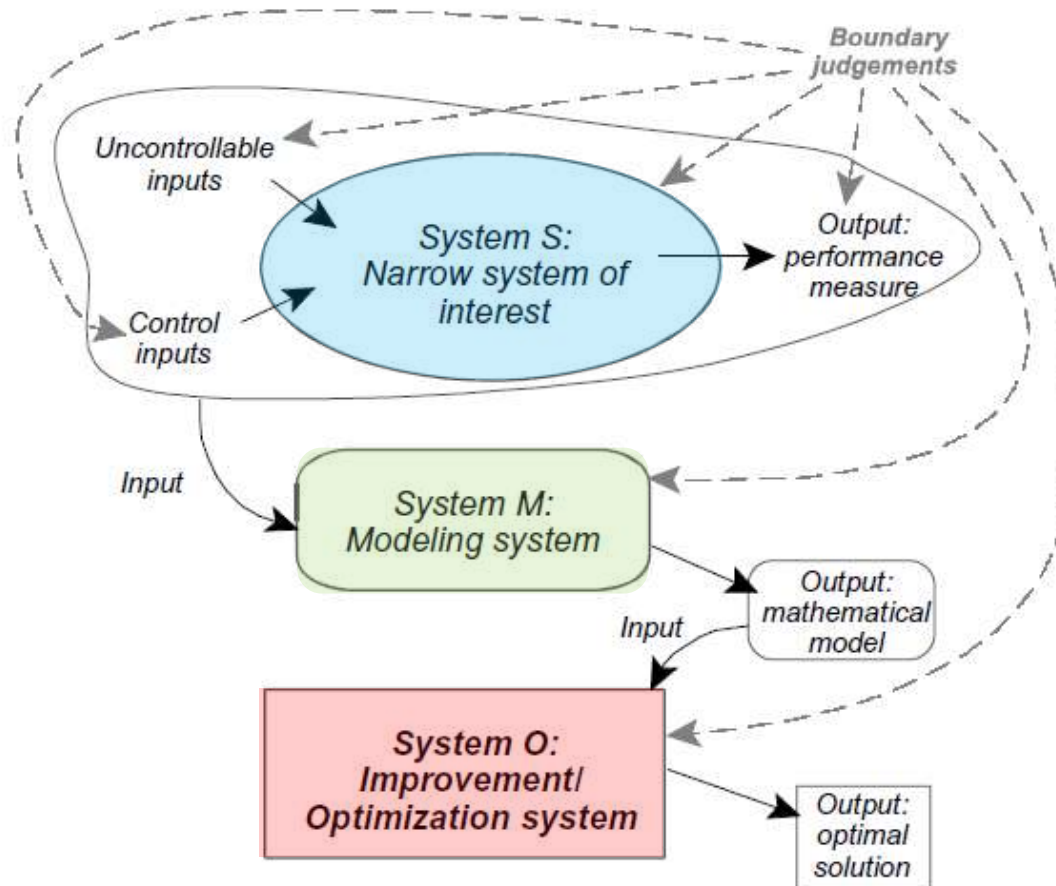
Source: Daellenbach (2005)



5. PROBLEM STRUCTURING METHOD

Overview Sistem/Model:

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Sistem S:

- Sistem yang menjadi interes.

Sistem M:

- Model matematik dari **Sistem S**.
- Dapat digunakan untuk menguji berbagai rencana pengadaan (*replenishment*) persediaan dan juga dampak perubahan pada input di luar kendali.

Sistem O:

- Mempunyai Sistem M sebagai input dan menghasilkan solusi yang lebih baik (*improved solution*).

Source: Daellenbach (2005)

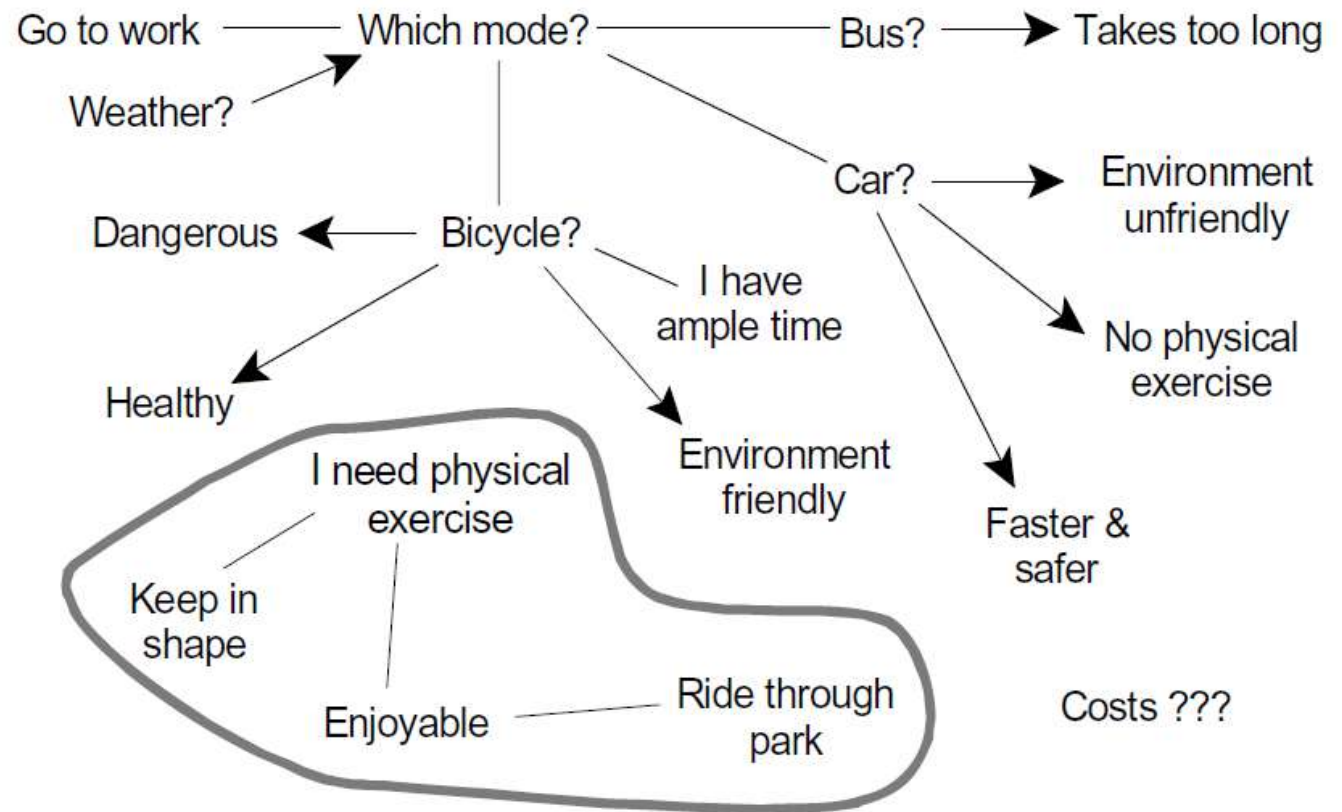


6. PROBLEM SITUATION SUMMARY

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(1) Mind maps

Diagrammatic aids to visualize a thought of a phenomenon, an issue, or a problem evoked including things, aspects, and concepts, also fears and aims, data and facts, and the possible actions and reactions by people or entities involved and their consequences, both planned and unplanned, desirable and undesirable.



Mind map of dilemma for going to work by bicycle

(Daellenbach and McNickle (2005), pp. 60)

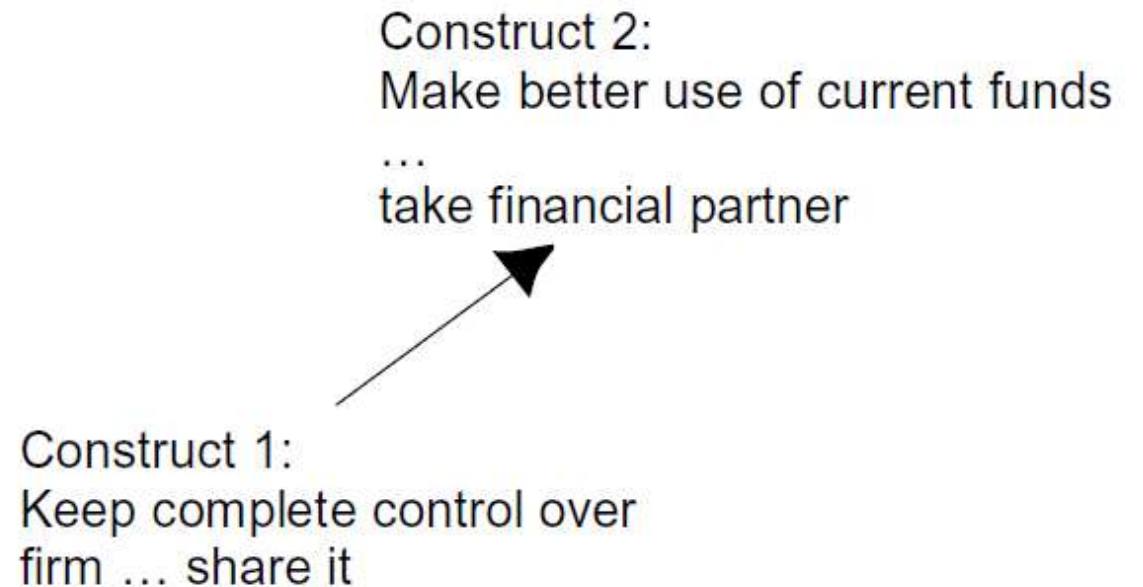


6. PROBLEM SITUATION SUMMARY

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(2) Cognitive mapping

- A tool that was adapted by C.L. Eden [1983] from G.A. Kelly's (1955) personal construct theory.
- A cognitive map only captures the subjective, personal perception of an individual.
- It takes the form of a network of statements, expressing concepts — ideas, goals, concerns, preferences, actions — and their contrasts or opposites.
- The concepts are linked together by arrows, which indicate the direction of connections



Constructs and links in cognitive mapping
(Daellenbach and McNickle (2005), pp. 66)



The diagram consists of 23 nodes, each representing a business event or decision. The nodes are interconnected by arrows indicating causal relationships. Some arrows are labeled with a '+' or '-' sign to indicate the polarity of the relationship. The nodes are as follows:

- [1] Funky shoes instant success ... a flop
- [2] Sales limited by production capacity ... demand
- [3] Cash position tight ... as good as predicted
- [4] 3-fold increase in RM stocks ... kept low
- [5] Locking up cash idle ... keeping funds free
- [6] Collections from retailers in 50 days ... 30 days
- [7] Potential sales easily doubled ... kept at current level
- [8] Get new suitable equipment now ... in 3 or 4 years
- [9] Generate \$140,000 for new equipment
- [10] Bank refuses loan without equal equity funds
- [11] No further private funds available
- [12] Get 50% financial partner plus bank loan ... do nothing
- [13] Nightmare come true
- [14] Keep complete control over firm ... share it
- [15] Competition fills market potential ... NuWave
- [16] Make better use of current funds ... take financial partner
- [17] Offer prompt payment discount ...
- [18] Reduces profits
- [19] Free enough funds for bank to offer balance ...
- [20] Operate efficiently with less RM stock ...
- [21] Risks delay in shipments ...
- [22] Reputation as unreliable supplier ...
- [23] Negative impact on sales ...

The diagram shows several key causal loops:

- Reinforcing Loop (R1):** [1] → [2] → [7] → [8] → [9] → [19] → [17] → [16] → [15] → [7]. This loop shows how initial success leads to increased sales, which then leads to equipment acquisition and financial decisions that further increase sales.
- Reinforcing Loop (R2):** [4] → [5] → [3] → [10] → [12] → [13] → [22] → [23] → [18] → [16] → [15] → [7]. This loop shows how low RM stocks lead to cash being locked up, which tightens the cash position and leads to financial decisions that eventually reduce profits and affect sales.
- Reinforcing Loop (R3):** [16] → [20] → [21] → [22] → [23] → [18] → [16]. This loop shows how better use of funds leads to operating efficiently, which risks delays and affects reputation, ultimately reducing profits.
- Reinforcing Loop (R4):** [16] → [14] → [8] → [9] → [19] → [17] → [16]. This loop shows how keeping control over the firm leads to equipment acquisition and financial decisions that further improve the firm's position.
- Reinforcing Loop (R5):** [16] → [17] → [19] → [17] → [16]. This loop shows how offering a payment discount leads to free funds for the bank, which then leads to better use of funds.
- Reinforcing Loop (R6):** [16] → [17] → [19] → [17] → [16]. This loop shows how offering a payment discount leads to free funds for the bank, which then leads to better use of funds.
- Reinforcing Loop (R7):** [16] → [17] → [19] → [17] → [16]. This loop shows how offering a payment discount leads to free funds for the bank, which then leads to better use of funds.
- Reinforcing Loop (R8):** [16] → [17] → [19] → [17] → [16]. This loop shows how offering a payment discount leads to free funds for the bank, which then leads to better use of funds.
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- Reinforcing Loop (R10):** [16] → [17] → [19] → [17] → [16]. This loop shows how offering a payment discount leads to free funds for the bank, which then leads to better use of funds.
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- Reinforcing Loop (R51):** [16] → [17] → [19] → [17] → [16]. This loop shows how offering a payment discount leads to free funds for the bank, which then leads to better use of funds.
- Reinforcing Loop (R52):** [16] → [17] → [19] → [17]



- **A cartoon-like pictorial summary of everything** (or almost everything!) the observer knows about the situation studied.
- Suggested by P. Checkland in SSM.



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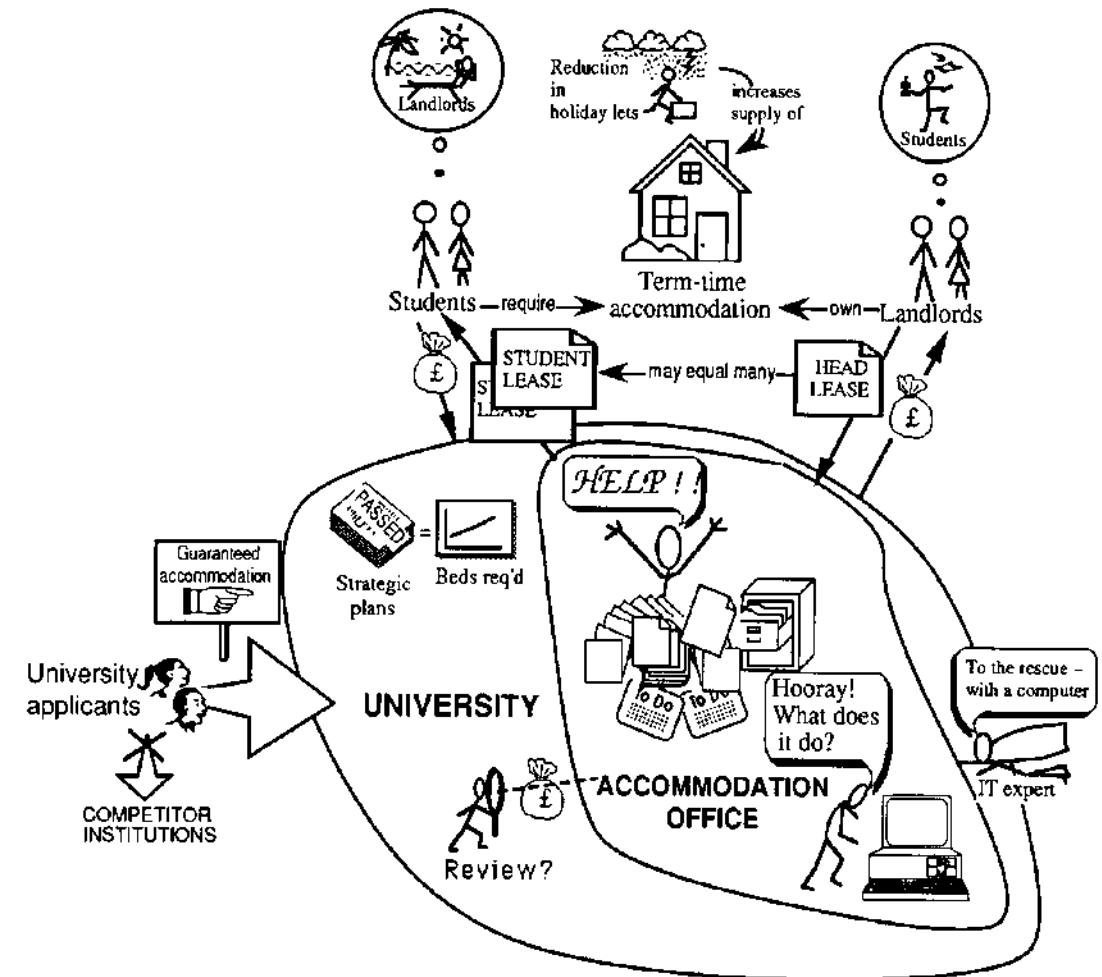
6. PROBLEM SITUATION SUMMARY

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- Ringkasan dalam bentuk gambar/ kartun tentang segala sesuatu yang diketahui pengamat/ analis
- Proses serupa dengan brain storming yang divisualisasikan
- Dibaca dari semua arah, dapat secara paralel (tidak harus serial)
- **Tidak pernah selesai**
- Bukan deskripsi sistem, melainkan SITUASI PROBLEMATIK

Situasi masalah: **akomodasi mahasiswa suatu universitas.**

- Pada titik pusat: Kantor akomodasi yang sangat disibukkan dengan *paperwork* – diperkirakan **komputerisasi** dapat menjadi jawaban.
- Daya saing universitas: **menjamin ketersediaan tempat**, tetapi jumlah peminat meningkat.
- Garis lingkaran: sistem nominal yang dipertimbangkan – **Sistem kantor akomodasi bagian dari sistem universitas yang lebih luas.**
- Beberapa tipe persewaan digambarkan bersama-sama berdasarkan persepsi mahasiswa dan landlords.

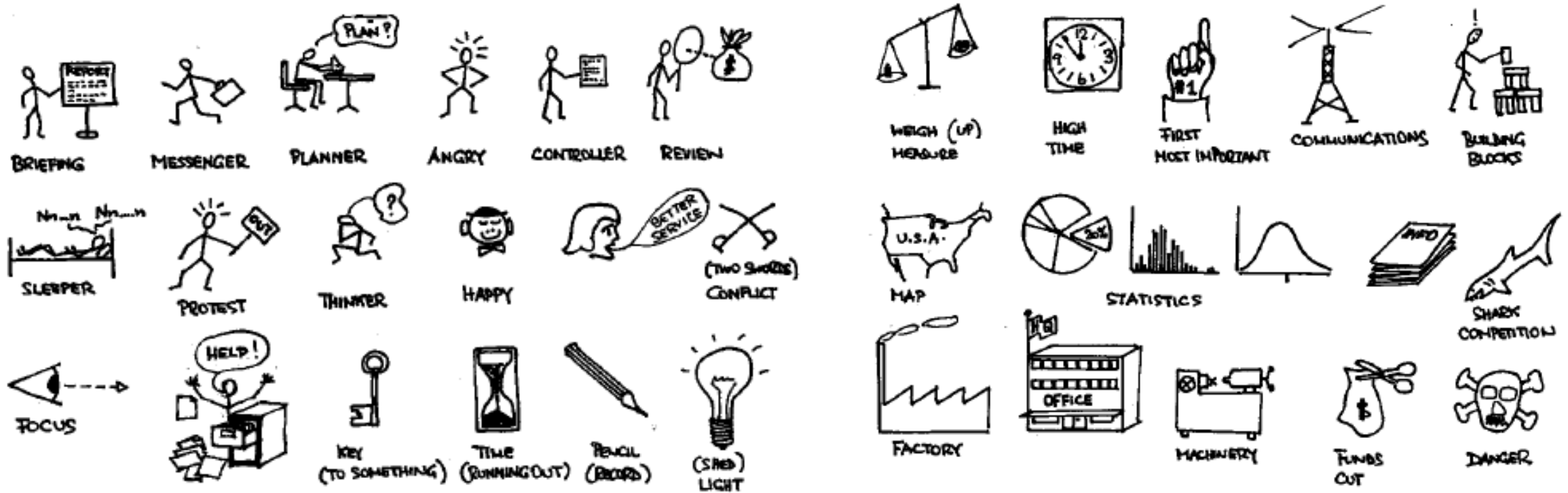




6. PROBLEM SITUATION SUMMARY

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Sample symbols for rich pictures.



Daellenbach and McNickle (2005), pp. 62.



6. PROBLEM SITUATION SUMMARY

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Komponen Utama RPD:

1. Elemen dari struktur:

- Semua aspek/komponen dari situasi yang relatif stabil (berubah tapi perlahan)
- Semua aspek dan elemen fisik
- Struktur fisik, bangunan & peralatan, produk, aspek, logika, fungsional, intelektual, seperti sifat komponen fisik & logika, bagian/departemen, informasi & data, aturan, dsb.

2. Elemen dari proses:

- Semua aspek situasi yang mengalami perubahan atau dalam kondisi mengalir
 - Contoh: Aktivitas yang berlangsung dalam proses
 - Aliran & proses material/informasi
 - Keputusan
 - yang diambil

3. Hubungan antara struktur & proses & antar proses:

- Bagian struktur mempengaruhi proses
- Bagian suatu proses mempengaruhi proses lain
- Output/aspek apa yang secara langsung atau tidak langsung dihasilkan dari proses tsb..



6. PROBLEM SITUATION SUMMARY

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Notes:

For **human activity systems**, a mind map or rich picture should include **‘hard’ facts** and **‘soft’ facts**.

1. Hard Facts (objective):

- Struktur fisik dan proses
- Catatan data, interpretasi statistikal, *information links*, segala sesuatu yang bersifat obyektif

2. Soft Facts (subjective):

- Opini, gosip, dugaan (hunches), hubungan interpersonal (pertemanan, konflik, kekuasaan, ego), *perceived agenda*, sinergi, dan hubungan simbiosis.
- Secara umum disebut “iklim” atau “suasana” dari situasi → determinan penting dari berbagai WORLD VIEW yang digunakan oleh orang yang terlibat dalam situasi tersebut.

Penggunaan RPD:

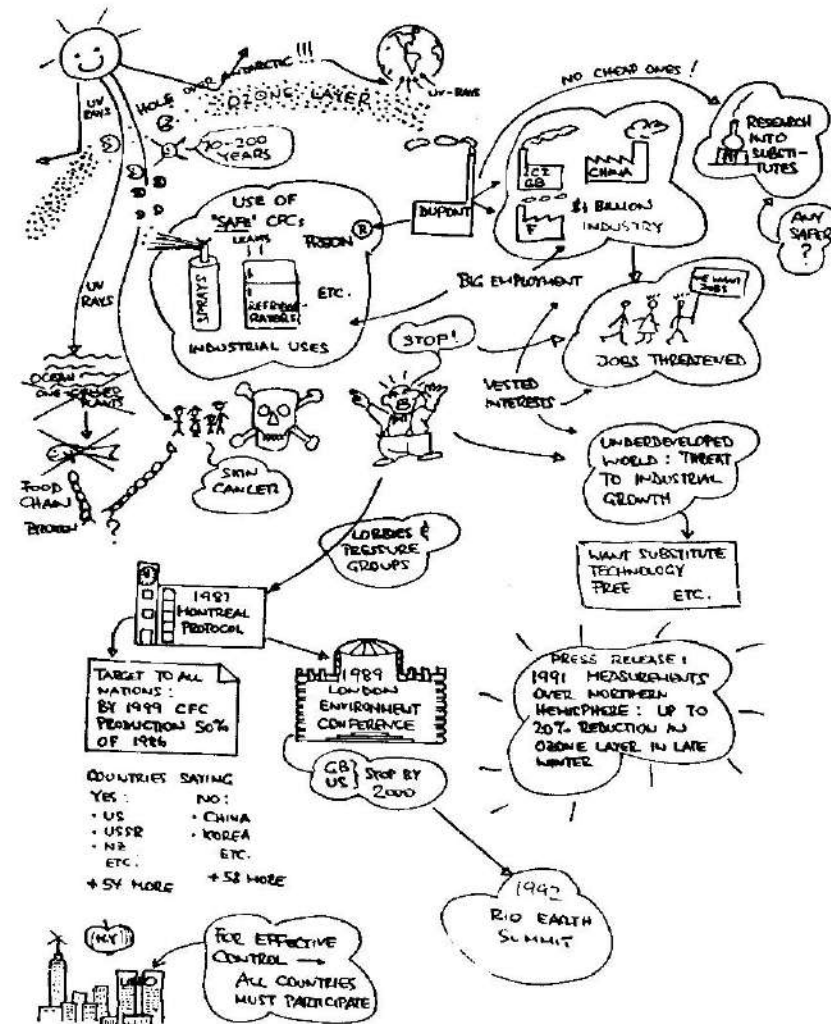
Alat bantu ideal untuk komunikasi dengan orang lain tentang situasi yang kompleks dan problematik.



6. PROBLEM SITUATION SUMMARY

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Contoh RPD: Masalah Lubang Ozon





6. PROBLEM SITUATION SUMMARY

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Kesalahan dalam Pembuatan RPD:

1. Setiap item dalam gambar harus terhubung satu sama lain → struktur dipaksa menjadi kompleks.
2. Indikasi RPD terlalu dipaksakan strukturnya adalah jika RPD menyerupai:
 - Aliran dokuman, informasi, material.
 - Diagram urutan (*precedence diagram*) dari suatu proses
 - Peta aliran (*flow chart*) dari proses keputusan



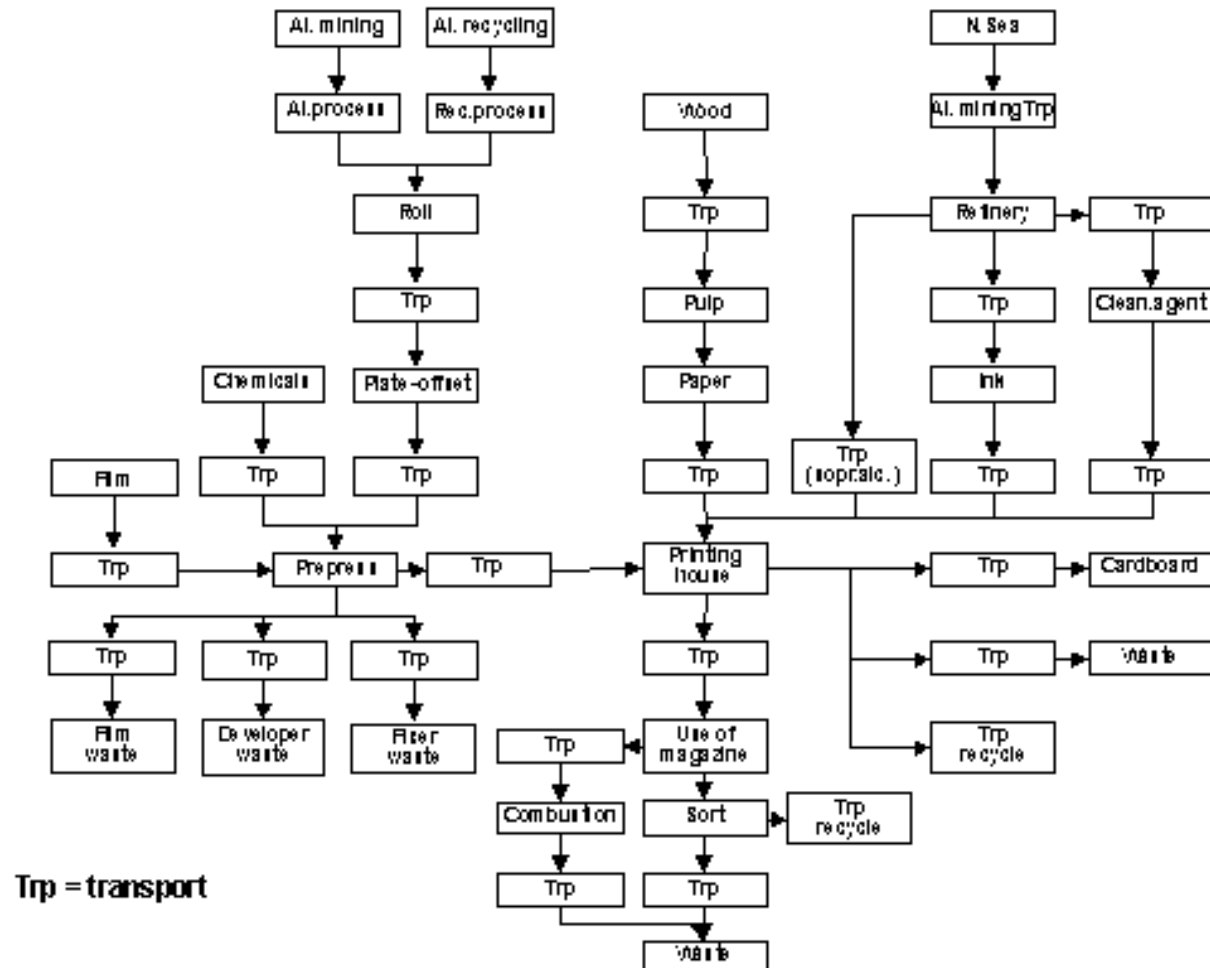
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Kesalahan dalam Pembuatan RPD:

- Aliran material

“Life-cycle-analysis” of magazine printing (materials)



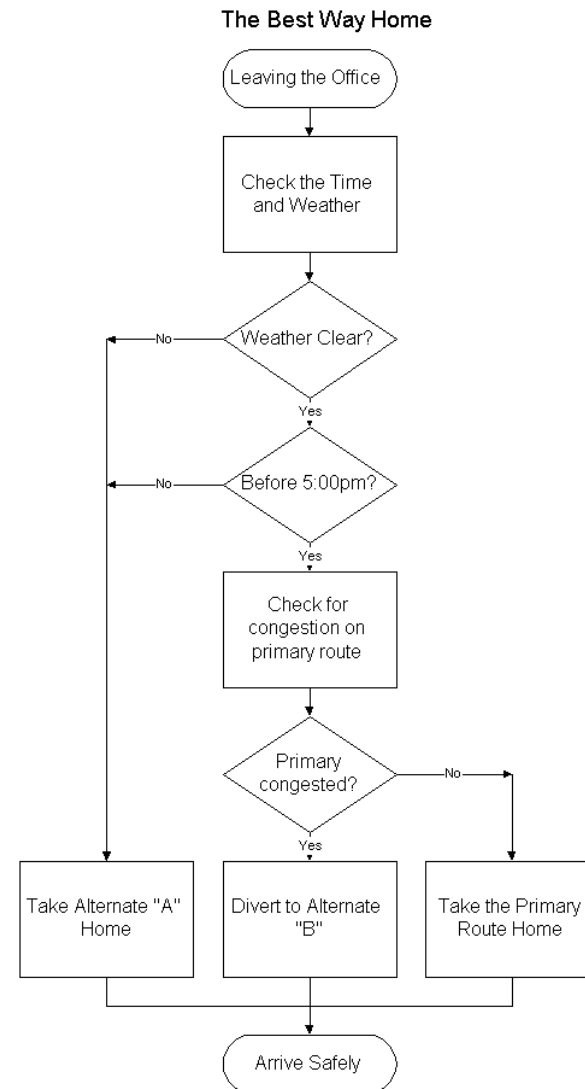


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Kesalahan dalam Pembuatan RPD:

- Peta Alir (*flow chart*)



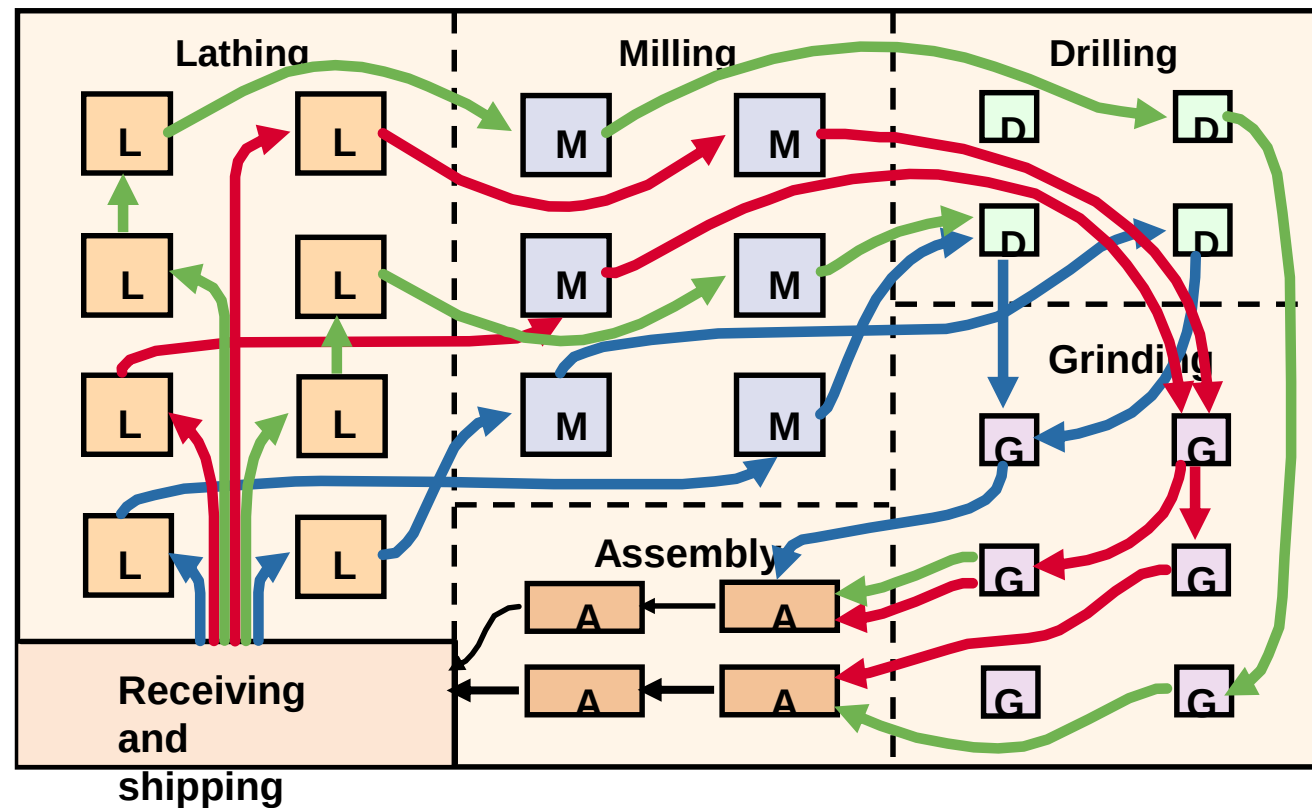


6. PROBLEM SITUATION SUMMARY

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Kesalahan dalam Pembuatan RPD:

- Aliran Proses



Jumbled flows in a job shop without GT (Group Technology) cells



6. PROBLEM SITUATION SUMMARY

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Properties of PSMs (Problem Structuring Methods) and Soft Systems Approaches

1. **Structuring the issues** in a problem situation, rather than narrowly focussed problem solving.
2. **Facilitating dialogue between the various stakeholders** with the aim of achieving
 - a greater degree of shared perceptions of the problem situation, rather than providing;
 - a decision aid to decision makers.
3. Addressing “What” questions first and only then “How” questions, i.e.
 - What is the **nature of the issue**?
 - What are **appropriate objectives**, given the various world views of the stakeholders?
 - What is the **appropriate definition of the system** for the issue considered?
 - **Which changes** are **systemically desirable** and **culturally feasible**?
 - **How are these changes** best brought about?



6. PROBLEM SITUATION SUMMARY

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Properties of PSMs (Problem Structuring Methods) and Soft Systems Approaches

4. Eliciting (memunculkan) the resolution of the problem from the stakeholders themselves, rather than from the analyst.
5. Changing the role of the problem solver to one of becoming a **facilitator and resource person** who relies on the technical subject expertise of the stakeholders.



7. CHECKLAND'S SOFT SYSTEMS METHODOLOGY

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Soft Systems Methodology (SSM)

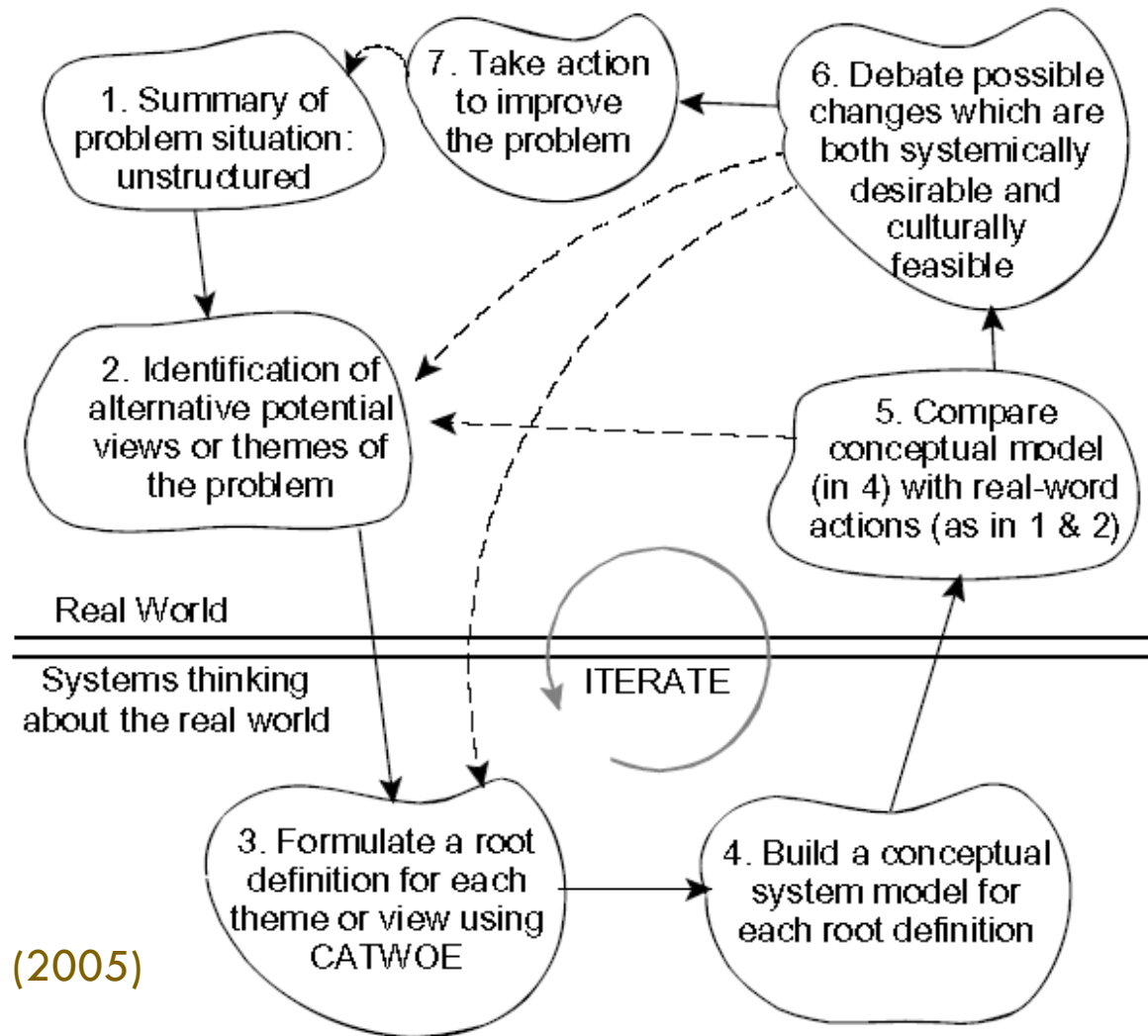
- SSM is a systemic **process of inquiry** which also happens to make use of systems models.
- 'Holon' is the **constructed abstract wholes** (instead using the word 'system')
- SSM uses a particular kind of holon: 'human activity system': a set of activities so connected as to make a purposeful whole, constructed to meet the requirement of the core system image



7. CHECKLAND'S SOFT SYSTEMS METHODOLOGY

Seven-stage Process Of Checkland's Soft Systems Methodology

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Source: Daellenbach (2005)



7. CHECKLAND'S SOFT SYSTEMS METHODOLOGY

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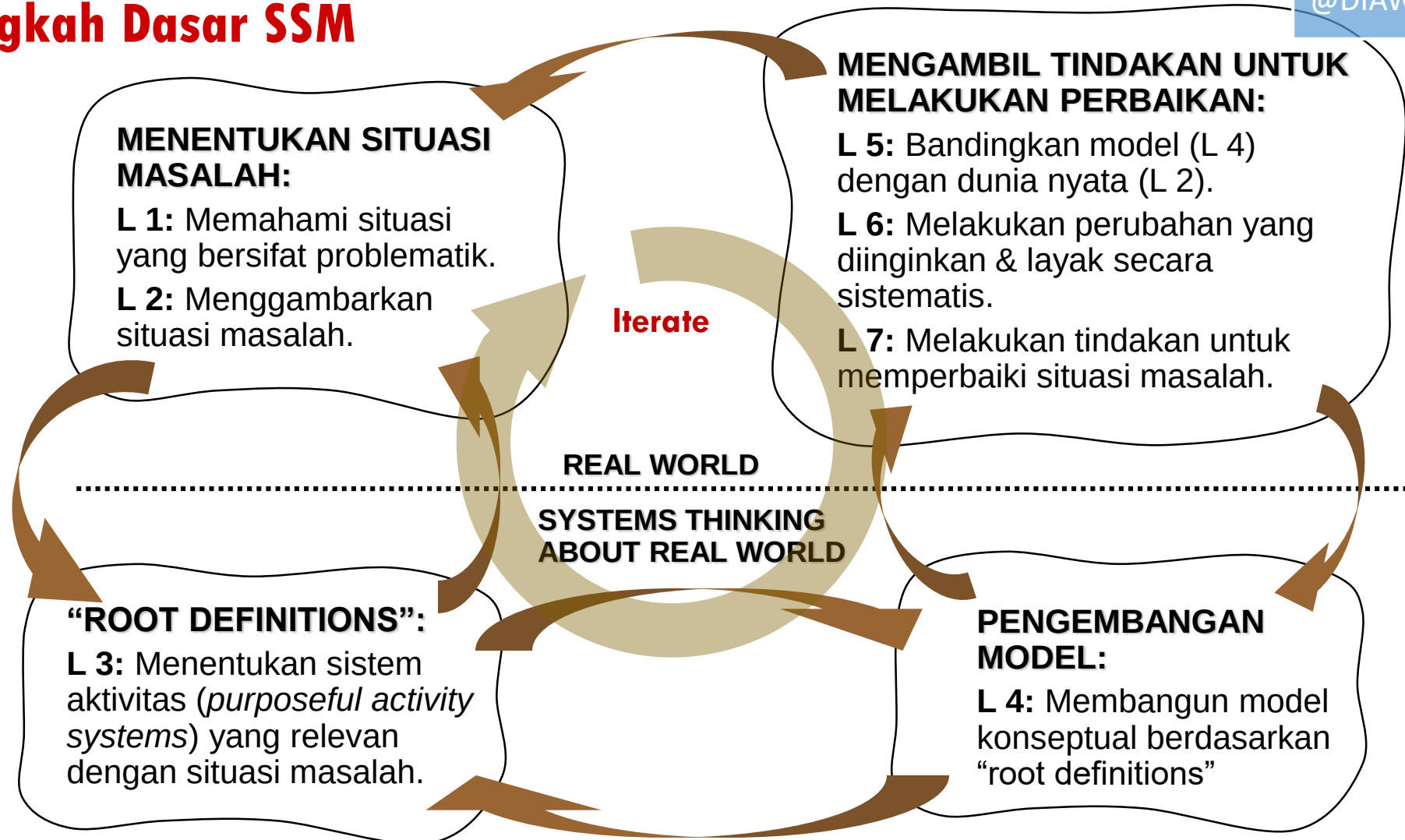
1. **Heart of SSM**: a comparison between the **world as it is**, and **some models of the world as it might be**.
2. **Interesting point**: SSM **constrains your thinking** in order for you to **expand your thinking**!
3. **Karakteristik SSM**:
 - *SSM: an **approach to inquiry into problem situations** perceived to exist in the real world (Checkland and Scholes, 1990).*
 - *SSM: a framework which **does not force or lead the systems analyst to a particular 'solution'**, rather to an **understanding** (Hirschheim et al., 1995).*
 - *SSM: is **situation-oriented**, not methodology-oriented.*
 - *SSM: is a **learning, not optimizing** system: learning has to be participative, so that SSM is not – or should not be – the skill of an external expert.*



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Tujuh Langkah Dasar SSM

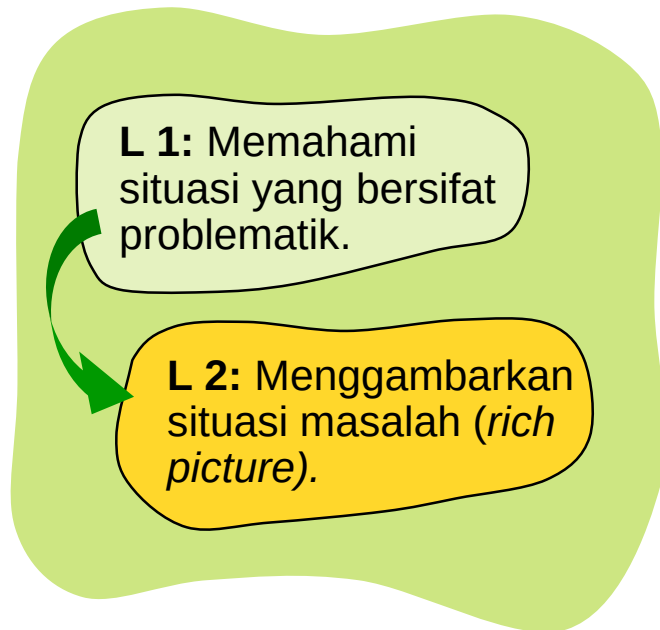




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L 1 & 2: Pendefinisian Situasi



L 2 Penggambaran situasi masalah:

- Menggambarkan situasi masalah; disarankan dengan menggunakan gambar: ***rich picture*** (Checkland, 1990).
- Elemen yang harus tercakup dalam *rich picture*:
 - ✓ Struktur
 - ✓ Proses
 - ✓ Iklim
 - ✓ Manusia
 - ✓ Isu, diekspresikan oleh manusia
 - ✓ Konflik



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Rich Picture (Checkland, 1990)

- A cartoon-like representation that identifies all the stakeholders, their concerns, and some of the structure underlying the work context (Monk and Howard, 1998).
- A pictorial summary of the actual situation in the “systems world” based on inquiries or observations of the “real world” (Patching, 1990).
- Represent the following
 - **Structure**
 - **Process**
 - **Concerns**



7. CHECKLAND'S SOFT SYSTEMS METHODOLOGY

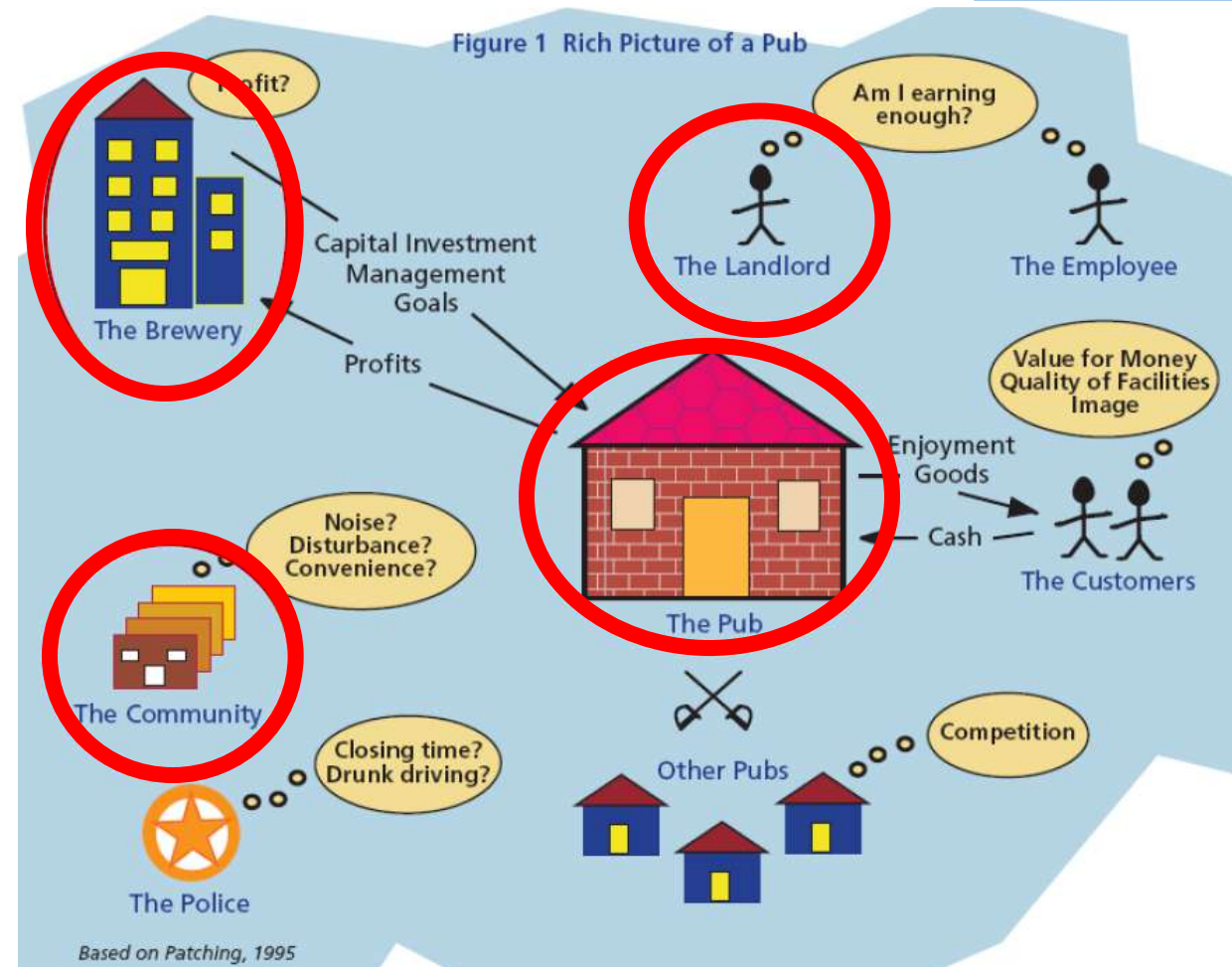
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Structure:

Aspect of the work context that are slow to change.

Such as,

- Stages of construction
- Firms/ organisations
- Geographical localities
- Stakeholders
- People





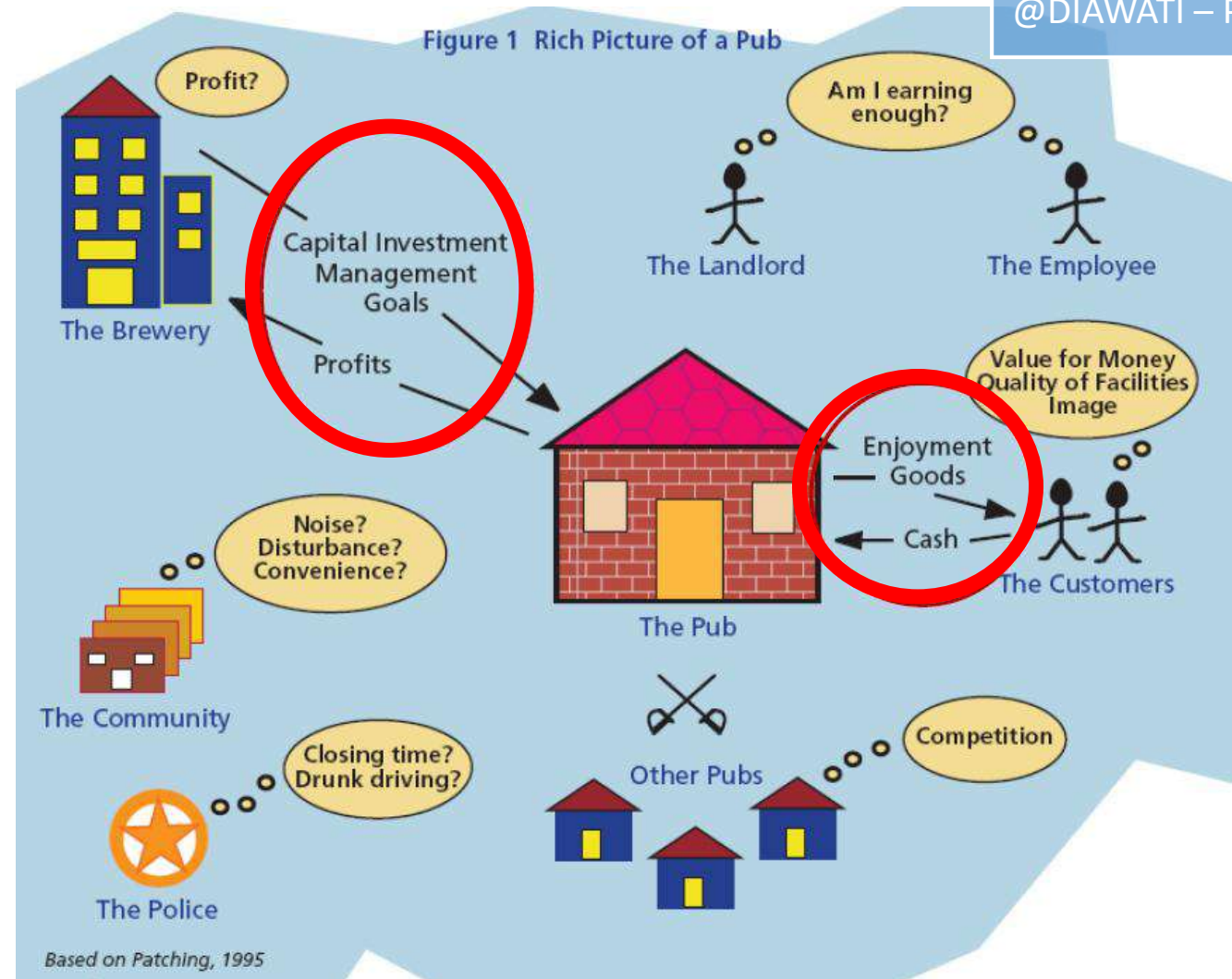
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Process:

Refers to the transformations that occur in the process of the work. Such as the flow of:

- Information/data
- Goods and services
- Other resources



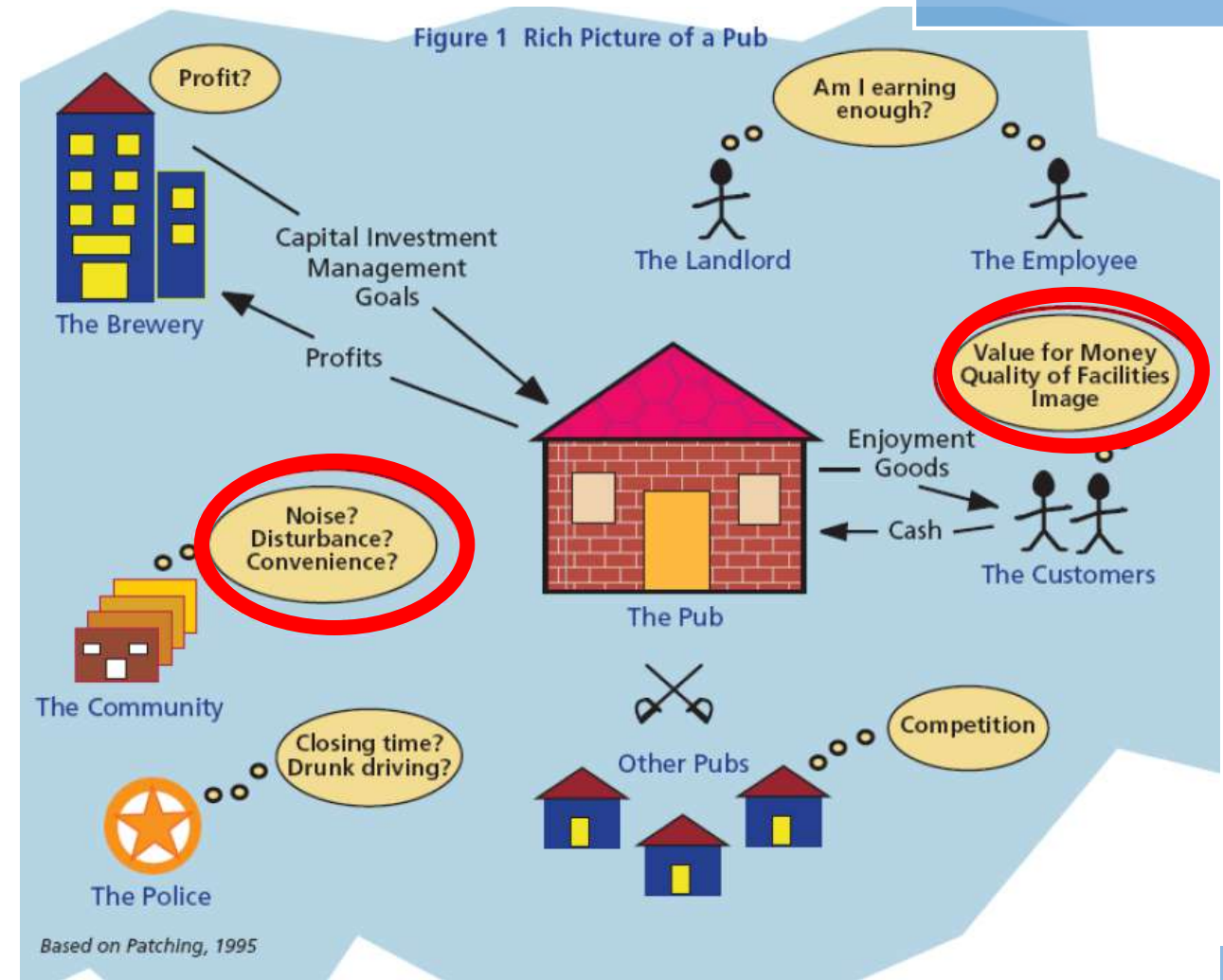


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Concerns:

- Perspectives/viewpoints of each individual's/firm's motivation
- Captures different perspectives of the theme
- Identifies tensions between stakeholders, a useful preliminary step to identifying the conflicting concerns and how they may be resolved.

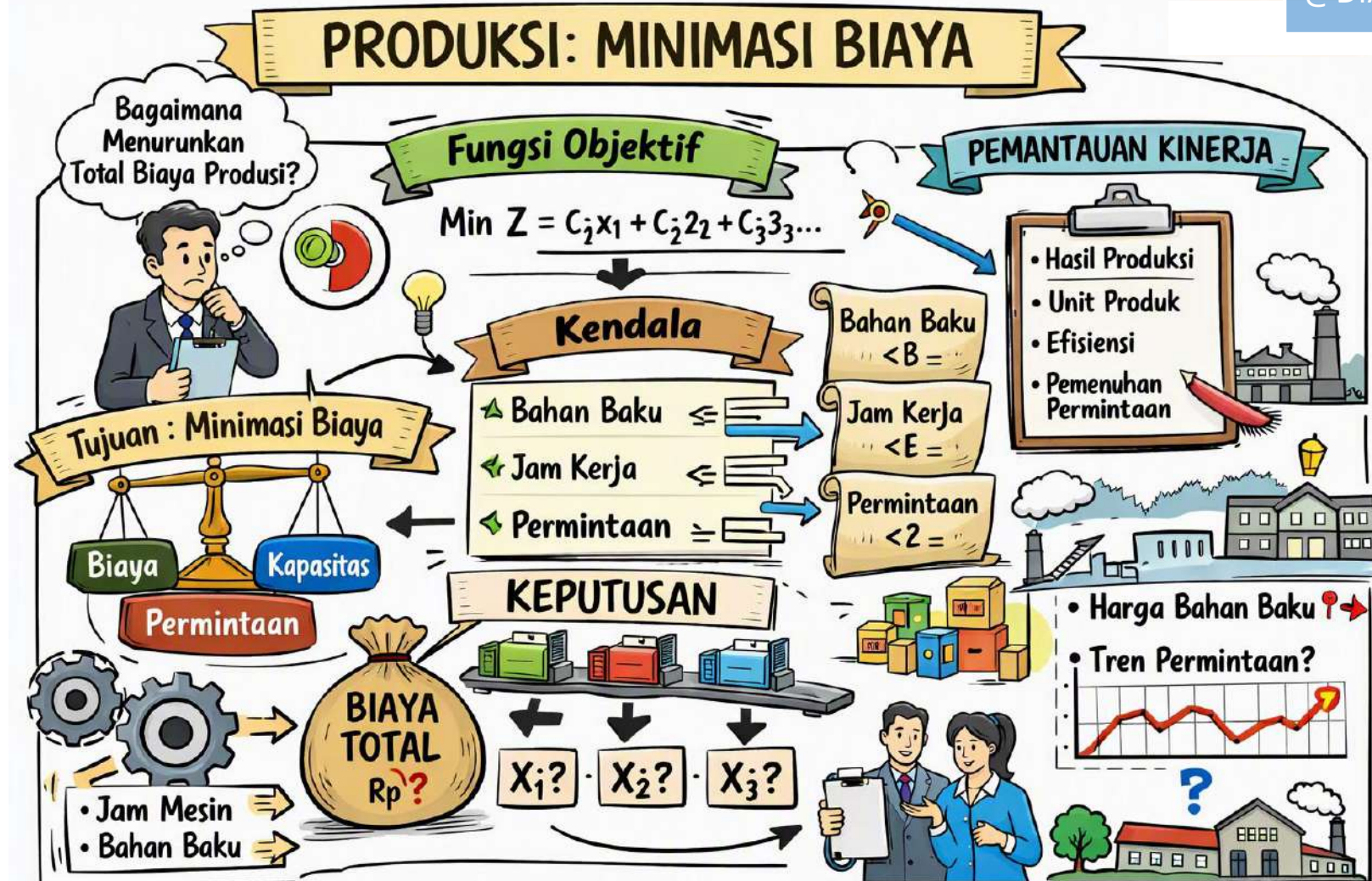




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RPD:
Minimasi Biaya
Produksi





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Dos and don'ts in making Rich Picture:

ELEMENT	EXPLANATION
Include <i>structure</i>	Include only enough structure to allow you to record the process and concerns.
Include <i>process</i>	Do not attempt to record all the details of process; a broad brush approach is usually all that is needed
Include <i>concerns</i>	Present the concern in a thought bubble
Use any pictorial or textual device that suits	There is no correct way of drawing a rich picture. There are as many styles as analysts and the same analyst will find different styles useful in different Situations



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“ROOT DEFINITIONS”:

L 3: Menentukan sistem aktivitas (*purposeful activity systems*) yang relevan dengan situasi masalah.

L 3 “Root Definitions”:

- Pengenalan, eksplorasi & pendefinisian situasi masalah dengan cara tertentu.
- Harus merupakan **representasi ringkas dari sistem** dalam bentuk yang paling dasar.
- Merefleksikan **tujuan inti** dari sistem aktivitas yang bertujuan dari persepsi yang dimodelkan.
- **Versi paling sederhana** dari “root definition”:
“*A system to do X by Y to achieve Z*”
- **Elemen** dalam “root definition”: **CATWOE**

CATWOE:

- **Customer:** yang mendapatkan dampak dari Transformasi.
- **Actor:** orang yang melakukan Transformasi.
- **Transformation:** proses merubah input menjadi output.
- **Weltanschauung:** sudut pandang, kerangka kerja atau image yang membuat “root definition” mempunyai makna.
- **Owner:** pihak yang memiliki kepentingan terbesar terhadap sistem & memiliki otoritas tertinggi untuk meniadakan keberadaan sistem tersebut.
- **Environment:** elemen dari luar sistem yang dapat mempengaruhi tetapi tidak mengendalikan sistem tersebut.



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L 3: “Root Definitions” dari sistem yang relevan

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Root definitions can either be:

- **Issue-based**, placing emphasis on the prime issue(s) the system is supposed to deal with, such as ‘resolve conflicts in resource allocations to different uses’, or
- **Primary-task** based, specifying the primary tasks that need to be executed, such as ‘allocate and efficiently use resources for different uses’, or a mixture.

Contoh:

Root definition 1: “A system controlled by Elly (O + A) for increasing production capacity (W), achieved through the purchase of new equipment (T), financed via freeing up funds by tight credit control and offering prompt payment incentives (T) to retailers (C), and by reducing current raw material stock levels, the latter by having the production supervisor (A) institute new purchasing policies (T) from suppliers (C).”

Root definition 2: “A system which allows Elly to convince the bank to advance sufficient additional funds for the purchase of new production equipment needed to capture the potential market for funky shoes, while keeping financial control over the firm.”



7. CHECKLAND'S SOFT SYSTEMS METHODOLOGY

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L 4: Pengembangan Model (conceptual models of the human activity systems)

PENGEMBANGAN MODEL:

L 4: Membangun model konseptual berdasarkan “root definitions”

L 4 Pengembangan Model:

- Pengembangan model berdasarkan “root definition” yang telah dibuat pada langkah sebelumnya.

■ Rekomendasi Checkland bagi pemula:

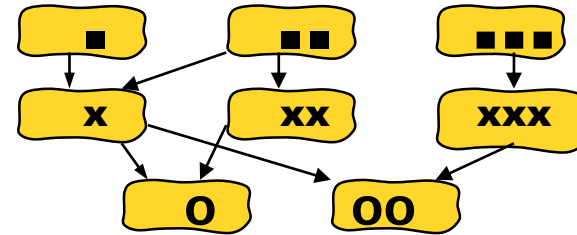
1. Tuliskan aktivitas yang diperlukan untuk melakukan proses Transformasi dengan menggunakan **kata kerja imperatif**. Target: **7 ± 2 aktivitas** pada skala yang sama.
2. Pilih aktivitas yang dapat dilakukan **secara bersamaan** (tidak saling bergantung satu sama lain).   
3. Tempatkan aktivitas-2 tersebut pada satu garis, kemudian tempatkan aktivitas yang bergantung pada aktivitas pertama tersebut pada satu garis, lanjutkan untuk seluruh aktivitas.



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4. Indikasikan **saling ketergantungan** antas aktivitas.



5. Susun ulang letak aktivitas untuk menghindari anak panah yang saling bersusun. Tambahkan cara untuk mengevaluasi kinerja & masukkan aspek lingkungan yang teridentifikasi dalam CATWOE.

6. Cek apakah model yang dikembangkan telah memenuhi **karakteristik sistem** sbb:

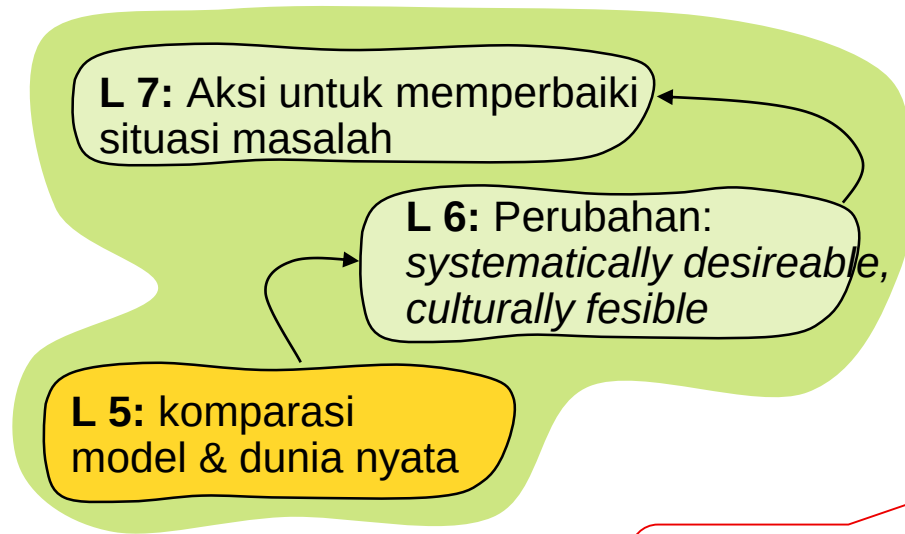
- *An ongoing purpose;*
- *A means of assessing performance;*
- *A decision taking process;*
- *Components that are also systems (i.e., the notion of sub-systems);*
- *Components that interact;*
- *An environment;*
- *A boundary between the system & the environment;*
- *Resources;*
- *Continuity.*



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L 5 - 7: Kembali ke Dunia Nyata



YANG PALING UMUM!

L 5 Komparasi model & dunia nyata:

■ Pendekatan:

1. Diskusi tak terstruktur;
2. Pertanyaan terstruktur dengan menggunakan **pendekatan matriks**;
3. Pemodelan skenario atau dinamis;
4. Uji coba memodelkan dunia nyata dengan struktur yang sama dengan model konseptual



7. CHECKLAND'S SOFT SYSTEMS METHODOLOGY

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Matriks Komparasi Antara Model Konseptual & Situasi Dunia Nyata

Aktivitas &
hubungan dari
model

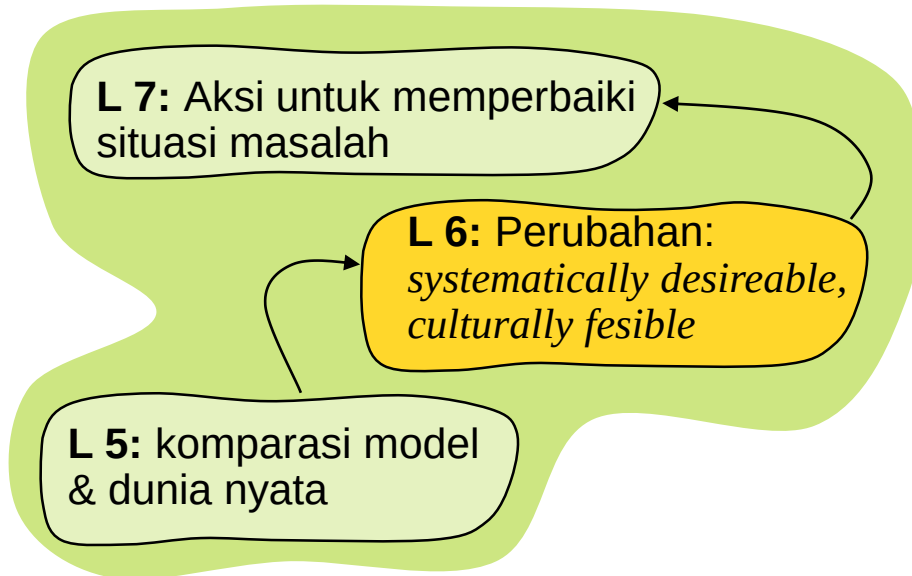
Aktivitas	Ada / tidak dalam situasi nyata	Bgmn hal ini dilakukan	Kriteria penilaian	Komentar: Ide untuk perubahan
1. -----				
2. -----				
3. -----				
Hubungan				
1 → 2				
⋮				
2 } → 4 3 }				



7. CHECKLAND'S SOFT SYSTEMS METHODOLOGY

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L 5 - 7: Kembali ke Dunia Nyata



L 6 Pengembangan intervensi yang diinginkan & layak:

- Pada langkah ini proses cenderung tidak sekuensial lagi, melainkan iteratif sepanjang 7 langkah untuk mendapatkan hasil perbaikan terbesar.
- Dilakukan eksplorasi intervensi yang paling mungkin berdasarkan analisis iteratif tersebut.
- **Penting (!):** evaluasi kelayakan intervensi merupakan aspek penting dalam SSM.



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Pendekatan untuk evaluasi kelayakan intervensi:

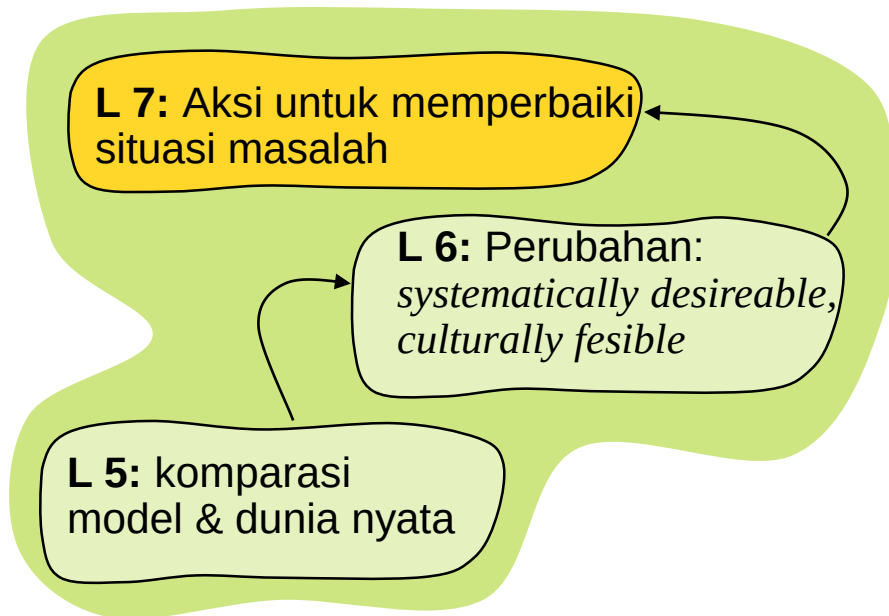
1. Lakukan pemodelan ulang dengan menggunakan CATWOE yang berbeda, perspektif yang berbeda, skala (sub-sistem model) yang berbeda;
2. Lakukan analisis berbasis sistem yang berbeda (*system dynamics, critical system thinking*);
3. Analisis “owner”: *“Siapa yang secara fundamental memiliki otoritas untuk melakukan tindakan?”*
4. “Analisis sistem sosial”: *“Bagaimana berbagai peran, norma & nilai yang ada dalam dunia nyata berhubungan dengan model konseptual?”*
5. “Analisis politikal”: *“Bagaimana kekuasaan diekspresikan dalam situasi yang dipelajari?”*



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L 5 - 7: Kembali ke Dunia Nyata



L 7 Aksi untuk perbaikan situasi :

- Merupakan **akhir satu siklus** atau menjadi **awal siklus baru**.
- Dapat menjadi situasi masalah tersendiri;
- Perubahan: ***systematically desirable*** (perubahan yang diinginkan & dilakukan secara sistematis), ***culturally fesible*** (kelayakan perubahan dapat tergantung pada budaya);
- Disiplin terkait: **manajemen perubahan**.



7. CHECKLAND'S SOFT SYSTEMS METHODOLOGY

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- **SSM sangat bermanfaat** pada saat diperlukan pemahaman yang kuat & mendalam dalam situasi masalah dengan:

1. Tujuan tidak tunggal;
2. Sudut pandang atau perspektif berbeda;
3. Asumsi berbeda;
4. Logika berbeda;
5. *Stakeholders* berbeda;
6. Sangat membingungkan.



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Pro & Kontra terhadap SSM:

Garis Kanan (PRO):

ASPEK	KRITIK	PENJELASAN
Teknikal	SSM tidak mempunyai metoda riel untuk cara membangun sistem	Banyak metodologi berbasis teknologi yang tidak berfungsi; yang diperlukan: menjaga komitmen & mempertimbangkan berbagai kepentingan.
Manajemen proyek	Karakteristik “ open ended ” pada SSM menyebabkan SSM tidak mungkin atau sukar dikelola.	Checkland: tidak ada cara untuk mengatakan apakah suatu proyek SSM berhasil atau tidak. Hal yang sama berlaku pula untuk pendekatan yang bersifat <i>prototyping</i> atau <i>evolutionary</i> . Mungkin: pemikiran traditional proyek manajemen tidak berlaku untuk perubahan organisasi.



7. CHECKLAND'S SOFT SYSTEMS METHODOLOGY

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Garis Kiri (KONTRA):

ASPEK	KRITIK	PENJELASAN
Asumsi SSM: seluruh anggota perusahaan mempunyai pilihan yang setara.	Asumsi berlebihan, diskusi terbuka antara atas & bawahan merupakan hal berlebihan. SSM mengabaikan isu kekuasaan atasan.	Langkah melakukan diskusi terbuka justru akan memicu perubahan pada budaya organisasi.
	Andai diskusi terbuka terjadi, sering berlaku " <i>common good</i> " view - " <i>what's good for the company is good for the workers</i> ".	Tindakan pencegahan yang mungkin: melibatkan reviewer / evaluator eksternal independen.
	Pada proyek SSM dapat terjadi manipulasi oleh konsultan, partisipan dapat dibuat untuk setuju dengan agenda tersembunyi konsultan	Tindakan pencegahan yang mungkin: menugaskan reviewer / evaluator internal atau evaluator eksternal independen.
Idealism SSM	SSM mengemukakan nilai keterbukaan & "kemanisan" (<i>niceness</i>) yang hanya sesuai untuk akademisi & bukan untuk manajer & pekerja.	SSM yang dikenalkan belum menyentuh kepentingan dasar masyarakat (situasi masalah riil).